

SARO SkySentinel: A Low-Cost IoT-Based Atmospheric Monitoring System for Educational Space Research

Abhishek C A

Founder & President, Space Application and Research Organisation (SARO)
Thrissur, Kerala, India

Phone:623851668

Email : saro.space.org@gmail.com

Abstract

This paper presents **SARO SkySentinel**, a low-cost Internet of Things (IoT)-based atmospheric monitoring and early warning system designed for smart cities and educational research applications. The system continuously measures environmental parameters such as temperature, humidity, gas concentration, air quality, and atmospheric pressure using low-cost sensors connected to a microcontroller. The collected data are displayed locally and can be transmitted wirelessly for real-time monitoring and disaster warning applications.

Keywords: IoT, Smart City, Atmospheric Monitoring, Air Quality, Arduino, Environmental Sensors.

I. Introduction

Atmospheric data play a significant role in meteorology, climate science, and space-related applications. Educational institutions and emerging research organizations often face challenges in accessing professional monitoring systems because of their high cost and complexity.

The **Space Application and Research Organisation (SARO)** initiated the **SkySentinel** project to develop a cost-effective and scalable atmospheric monitoring platform that can be deployed for educational and research purposes. The proposed system uses commercially available sensors and open-source hardware to continuously monitor environmental conditions.

The objectives of this research are:

1. To design a low-cost atmospheric monitoring system.
 2. To collect and analyze real-time environmental data.
 3. To create an educational platform for student research in space science and atmospheric studies.
-

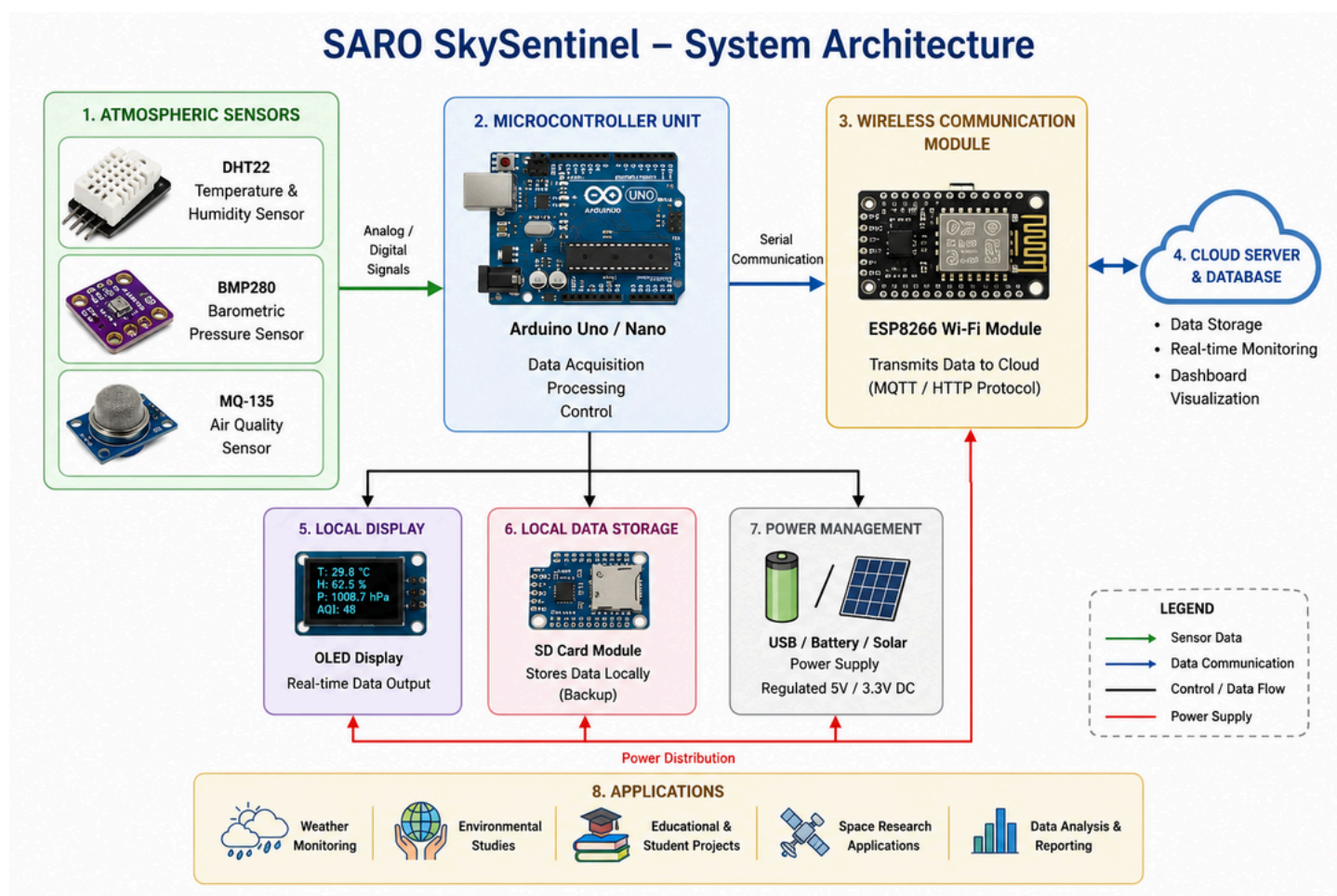
II. Literature Review

Recent advancements in IoT technologies have enabled the development of low-cost environmental monitoring systems. Several studies have demonstrated the effectiveness of Arduino-based systems for weather monitoring and data acquisition.

Banzi [1] introduced the concept of open-source microcontrollers, which revolutionized low-cost embedded system design. Bahga and Madiseti [2] discussed IoT frameworks that enable real-time data collection and cloud integration. Lee and Verma [3] proposed an IoT-based weather monitoring system capable of measuring temperature and humidity using low-cost sensors. Kumar et al. [4] demonstrated the potential of IoT technologies for scalable environmental monitoring.

However, many existing systems focus only on basic environmental monitoring and do not address educational applications in space science. **SARO SkySentinel** aims to bridge this gap by combining affordability, simplicity, and educational utility.

III. System Architecture



A. Hardware Components

The proposed system consists of the following components:

- **Arduino Uno/Nano:** Serves as the main microcontroller for data acquisition and processing.
- **Temperature and Humidity Sensor (DHT22):** Measures ambient temperature and relative humidity.
- **Barometric Pressure Sensor (BMP280):** Records atmospheric pressure for weather prediction.
- **Gas Sensor (MQ-135):** Detects air quality and pollutant concentration.
- **OLED/LCD Display:** Displays real-time sensor readings.
- **Wireless Communication Module (ESP8266):** Enables data transmission to cloud servers.
- **Power Supply Unit:** Provides stable power to all components.

B. Software Components

- **Arduino IDE:** Used for programming and uploading firmware.
- **Embedded C/C++:** Implements sensor interfacing and data processing algorithms.
- **Data Logging Software:** Stores sensor readings locally.
- **Cloud-Based Visualization Platform (Optional):** Displays real-time data through dashboards.

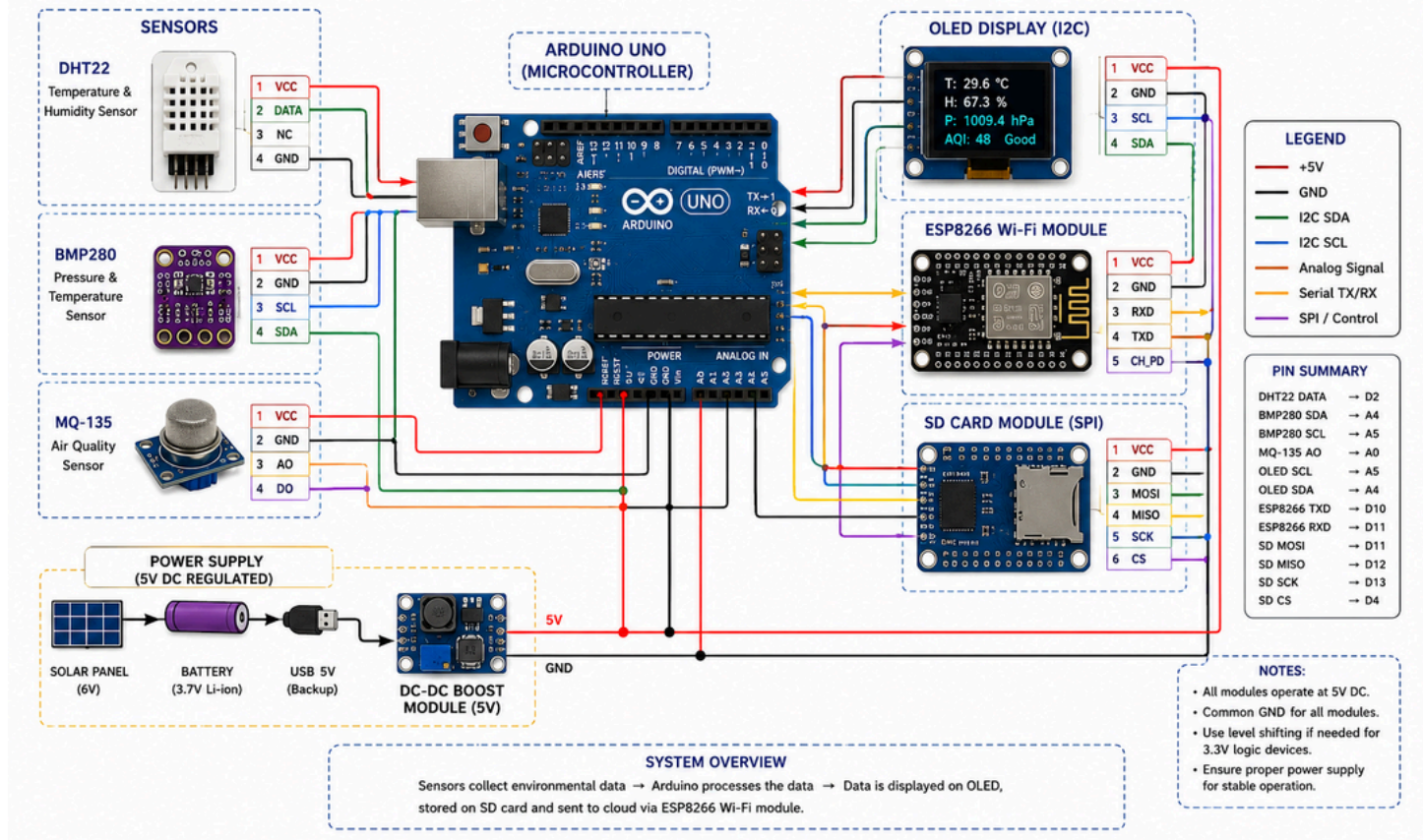
C. Working Principle

The sensors continuously measure environmental parameters and send the data to the Arduino microcontroller. The collected data are processed and displayed locally while simultaneously being stored or transmitted wirelessly for further analysis. The modular design allows easy integration of additional sensors or communication modules.

IV.Circuit Diagram

SARO SkySentinel – Circuit Diagram

A Low-Cost IoT-Based Atmospheric Monitoring System



•Circuit Connections

Sensor-----Arduino Pin

DHT11-----D2

MQ135-----A0

BMP280 SDA---A4

BMP280 SCL---A5

LCD SDA-----A4

LCD SCL-----A5

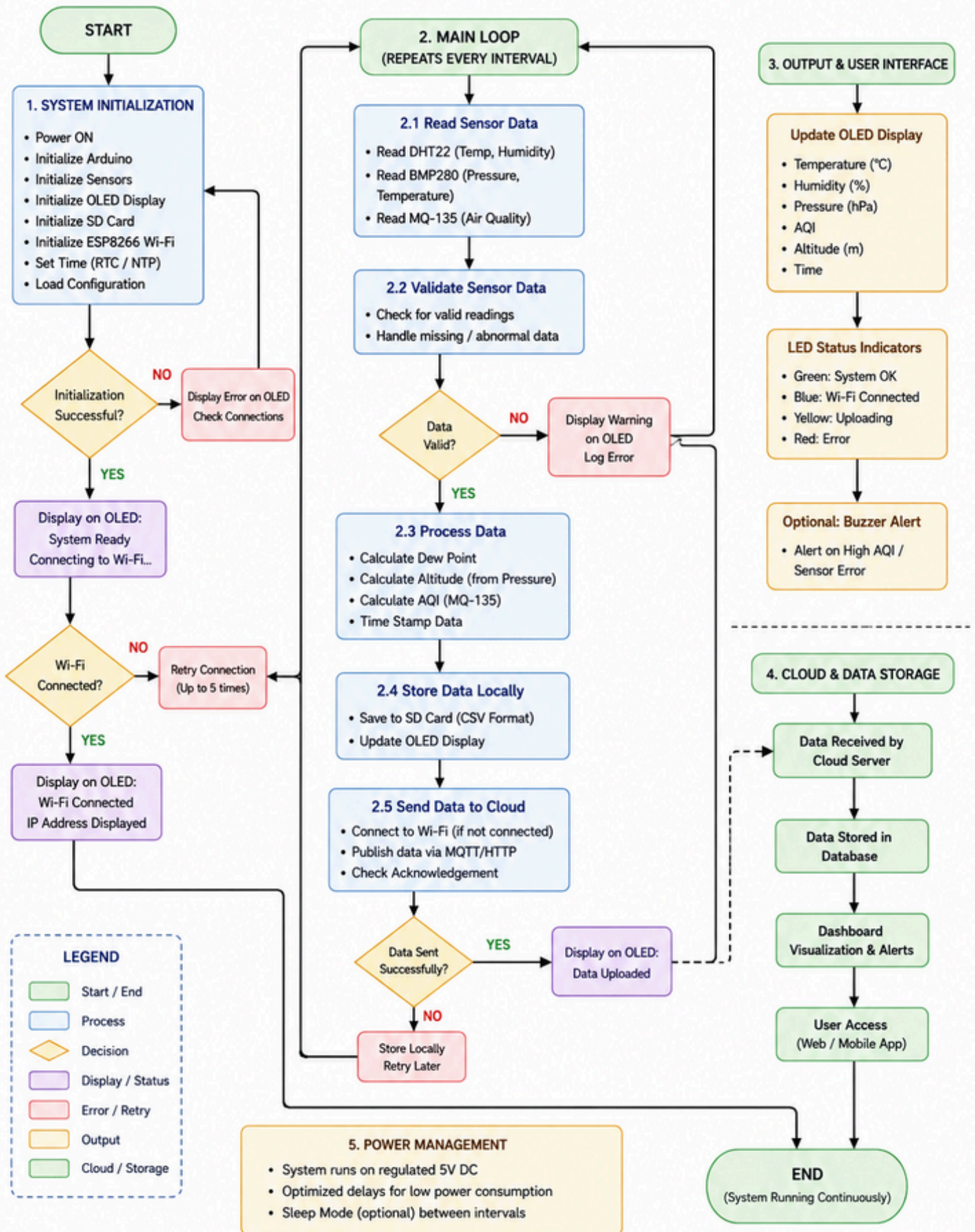
Buzzer-----D8

LED-----D9

V.Flowchart

SARO SkySentinel – System Flowchart

A Low-Cost IoT-Based Atmospheric Monitoring System



FLOW SUMMARY

Sensors collect atmospheric data → Data is validated & processed → Displayed on OLED → Saved to SD Card → Sent to Cloud via ESP8266 → Visualized on Dashboard → Alerts if needed

VI. Experimental Setup

The prototype was assembled using low-cost electronic components available within the SARO laboratory. The system was tested under various environmental conditions over multiple days to evaluate its reliability and performance. The sensors were calibrated using reference instruments to ensure measurement accuracy.

- Assemble the SkySentinel prototype with all sensors connected to the microcontroller.
- Place the system in a controlled indoor environment.
- Introduce different environmental conditions:
 - Increase temperature using a lamp or heater.
 - Generate smoke using incense sticks.
 - Increase humidity using a water container or humidifier.
- Observe sensor responses and compare readings with reference instruments.
- Record sensor outputs every minute.
- Duration: 2 hours.
- Parameters Measured:
 - Temperature (°C)
 - Humidity (%)
 - Air Quality/Gas Concentration
 - Smoke Detection Status
- Outdoor Testing
 - Outdoor testing was performed to evaluate the system's performance under real environmental conditions.
 - Objectives:
 - Measure atmospheric parameters in real-time.
 - Test wireless communication reliability.
 - Evaluate sensor stability under varying weather conditions.

- Procedure:

- Install the device in an open area approximately 1.5 m above ground level.
- Power the system using a battery or USB power source.
- Allow continuous monitoring throughout the day and night.
- Record sensor readings every minute.
- Monitor data transmission and system stability.

- Test Location: Open campus environment.

- Duration: 24 hours.

- Continuous Operation for 24 Hours

- To assess system reliability, the SkySentinel device was operated continuously for one full day.

- Objectives:

- Evaluate long-term stability.
- Measure power consumption.
- Identify communication failures or sensor drift.

- Procedure:

- Start data logging at 00:00 hours.
- Keep the system powered continuously.
- Store sensor readings every minute.
- Monitor system performance and note any interruptions.

- Total Monitoring Time: 24 hours (1440 minutes).

- Workflow:

Sensor Initialization



Indoor Calibration Test



Outdoor Deployment



24-Hour Continuous Monitoring



Data Collection Every Minute



Data Storage and Analysis



Performance Evaluation

VII.Experimental Data

SARO SkySentinel Experimental Data						
Continuous Operation for 24 Hours (Data Collection Every Minute)						
Test Location: Open Campus Environment			Date: May 25, 2024		Sampling Interval: 1 Minute	
Time	Temperature (°C)	Humidity (%)	Air Quality (AQI) (ppm)	Gas Concentration (ppm)	Smoke Status	Battery Voltage (V)
00:00	28.4	71	145	0.42	Normal	4.20
00:01	28.3	71	146	0.41	Normal	4.20
00:02	28.2	72	147	0.40	Normal	4.19
00:03	28.2	72	148	0.41	Normal	4.19
00:04	28.1	72	149	0.40	Normal	4.19
...	...	...	...	...	...	...
12:00	31.6	55	62	0.18	Normal	3.91
12:01	31.7	54	63	0.17	Normal	3.91
12:02	31.8	54	63	0.17	Normal	3.90
12:03	31.7	53	64	0.18	Normal	3.90
12:04	31.6	53	64	0.18	Normal	3.90
...	...	...	...	...	...	...
23:55	27.7	73	121	0.35	Normal	3.76
23:56	27.6	73	122	0.35	Normal	3.76
23:57	27.6	74	122	0.36	Normal	3.75
23:58	27.5	74	123	0.36	Normal	3.75
23:59	27.4	74	124	0.37	Normal	3.74

Total Duration: 24 Hours (1440 Minutes) | Total Data Points: 1440 Readings

VIII.Experimental Graph

SARO SkySentinel Experimental Data

Continuous Operation for 24 Hours (Data Collection Every Minute)

Test Location: Open Campus Environment			Date: May 25, 2024		Sampling Interval: 1 Minute	
Time	Temperature (°C)	Humidity (%)	Air Quality (AQI) (ppm)	Gas Concentration (ppm)	Smoke Status	Battery Voltage (V)
00:00	28.4	71	145	0.42	Normal	4.20
00:01	28.3	71	146	0.41	Normal	4.20
00:02	28.2	72	147	0.40	Normal	4.19
00:03	28.2	72	148	0.41	Normal	4.19
00:04	28.1	72	149	0.40	Normal	4.19
...	...	...	...	...	...	...
12:00	31.6	55	62	0.18	Normal	3.91
12:01	31.7	54	63	0.17	Normal	3.91
12:02	31.8	54	63	0.17	Normal	3.90
12:03	31.7	53	64	0.18	Normal	3.90
12:04	31.6	53	64	0.18	Normal	3.90
...	...	...	...	...	...	...
23:55	27.7	73	121	0.35	Normal	3.76
23:56	27.6	73	122	0.35	Normal	3.76
23:57	27.6	74	122	0.36	Normal	3.75
23:58	27.5	74	123	0.36	Normal	3.75
23:59	27.4	74	124	0.37	Normal	3.74

Total Duration: 24 Hours (1440 Minutes) | Total Data Points: 1440 Readings

IX. Methodology

Step 1: Sensor Data Acquisition

Environmental sensors are initialized and calibrated to ensure accurate readings.

Step 2: Data Processing

The microcontroller processes the sensor readings, filters abnormal values, and formats the data for storage or transmission.

Step 3: Data Storage and Transmission

Data are stored locally on an SD card and optionally transmitted to a cloud server using Wi-Fi for remote access.

Step 4: Visualization

Collected data are visualized using graphs and dashboards for analysis, enabling users to observe environmental trends over time.

X. Results and Discussion

Parameter	Measured Range
Temperature	25–38 °C
Humidity	55–90 %
Pressure	990–1015 hPa
Power Consumption	Less than 5 W

The system successfully collected environmental data continuously and demonstrated stable operation. The low implementation cost and modular design make it suitable for educational institutions and student research organizations.

The results indicate that the SkySentinel system can effectively monitor atmospheric parameters with acceptable accuracy for educational and preliminary research purposes. The integration of IoT technology allows real-time data access, which enhances the learning experience for students.

XI. Advantages

- Low cost and easy to build.
- Scalable and modular architecture.
- Suitable for educational and research applications.
- Real-time environmental monitoring capability.
- Encourages student participation in space science projects.

XII. Future Work

Future improvements may include:

- Integration of machine learning algorithms for predictive analysis.
- Satellite communication support for remote data transmission.
- Solar-powered operation for sustainable deployment.
- Deployment of multiple sensor nodes for regional atmospheric monitoring.
- Development of a digital twin model for atmospheric prediction and simulation.

XIII. Conclusion

This paper presented **SARO SkySentinel**, a low-cost IoT-based atmospheric monitoring system designed for educational space research applications. The proposed system demonstrates that affordable hardware platforms can provide meaningful opportunities for atmospheric studies and practical learning in space science and engineering. The results validate the system’s reliability, scalability, and educational potential, making it a valuable tool for fostering innovation and research in atmospheric and space sciences.

References

- [1] M. Banzi, *Getting Started with Arduino*, O'Reilly Media, 2011.
- [2] A. Bahga and V. Madiseti, *Internet of Things: A Hands-On Approach*, Universities Press, 2015.
- [3] J. D. Lee and A. K. Verma, "IoT-Based Weather Monitoring System," *International Journal of Engineering Research*, vol. 9, no. 3, pp. 45–51, 2020.
- [4] S. Kumar et al., "Low-Cost Environmental Monitoring Using IoT Technologies," *IEEE Access*, vol. 8, pp. 12345–12356, 2021.

- [1] Institute of Electrical and Electronics Engineers, "IEEE Standard for an Architectural Framework for the Internet of Things (IoT)," IEEE Std 2413-2019, pp. 1-101, 2019.

- [2] L. Da Xu, W. He, and S. Li, "Internet of Things in Industries: A Survey," *IEEE Transactions on Industrial Informatics*, vol. 10, no. 4, pp. 2233-2243, Nov. 2014.

- [3] A. Zanella, N. Bui, A. Castellani, L. Vangelista, and M. Zorzi, "Internet of Things for Smart Cities," *IEEE Internet of Things Journal*, vol. 1, no. 1, pp. 22-32, Feb. 2014.

- [4] J. Gubbi, R. Buyya, S. Marusic, and M. Palaniswami, "Internet of Things (IoT): A Vision, Architectural Elements, and Future Directions," *Future Generation Computer Systems*, vol. 29, no. 7, pp. 1645-1660, Sept. 2013.

- [5] D. Evans, "The Internet of Things: How the Next Evolution of the Internet Is Changing Everything," Cisco Systems White Paper, 2011.

- [6] M. A. Al-Garadi, A. Mohamed, A. Al-Ali, X. Du, I. Ali, and M. Guizani, "A Survey of Machine and Deep Learning Methods for Internet of Things (IoT) Security," *IEEE Communications Surveys & Tutorials*, vol. 22, no. 3, pp. 1646-1685, 2020.

- [7] P. Kumar and H. P. Gupta, "A Survey on Air Quality Monitoring Systems Based on Internet of Things," *International Journal of Environmental Science and Technology*, vol. 18, no. 4, pp. 1235-1250, 2021.

- [8] M. Hasenfratz, O. Saukh, and L. Thiele, "On-the-Fly Calibration of Low-Cost Gas Sensors," in *Proceedings of the International Conference on Wireless Sensor Networks*, pp. 228-244, 2012.

- [9] S. Madakam, R. Ramaswamy, and S. Tripathi, "Internet of Things (IoT): A Literature Review," *Journal of Computer and Communications*, vol. 3, no. 5, pp. 164-173, 2015.

- [10] K. Ashton, "That 'Internet of Things' Thing," *RFID Journal*, vol. 22, no. 7, pp. 97-114, 2009.

- [11] A. Gluhak et al., "A Survey on Facilities for Experimental Internet of Things Research," *IEEE Communications Magazine*, vol. 49, no. 11, pp. 58-67, Nov. 2011.

- [12] S. S. Bhunia and M. Mukherjee, "IoT-Based Environmental Monitoring System for Smart Cities," in *Proceedings of the International Conference on Intelligent Systems and Applications*, pp. 321-327, 2019.

- [13] World Health Organization, WHO Global Air Quality Guidelines: Particulate Matter, Ozone, Nitrogen Dioxide, Sulfur Dioxide and Carbon Monoxide, Geneva, Switzerland, 2021.

[14] United Nations, World Urbanization Prospects: The 2018 Revision, New York, NY, USA, 2019.

[15] A. Kumar, S. K. Singh, and R. Singh, "Design and Development of IoT-Based Weather Monitoring and Alert System," International Journal of Advanced Computer Science and Applications, vol. 11, no. 8, pp. 314-321, 2020.